

Adult Census Income: Analysis and Prediction

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Abstract

The Adult Census Income dataset is a widely used benchmark for socio-economic classification, containing demographic and employment-related attributes collected from the 1994 US Census Bureau database. The objective of this project is to predict whether an individual’s annual income exceeds **\$50,000** based on census features such as age, education, occupation, working hours, and capital gains.

We design and implement a complete machine learning workflow using **KNIME Analytics Platform**, following a strict *no-leakage architecture*. This ensures that preprocessing transformations, feature engineering steps, and sampling techniques are learned exclusively from the training partition and then applied to the test data without re-fitting. The pipeline includes data cleaning, imputation, Z-score normalization, dimensionality reduction through PCA, demographic clustering using k-means, SMOTE-based balancing, and evaluation of three classifiers: Logistic Regression, XGBoost, and Multi-Layer Perceptron.

Keywords

Machine Learning — Binary Classification — Census Data — Income Prediction — KNIME Workflow — XGBoost — SMOTE — ROC-AUC

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Project Repository: <https://github.com/smabbasht/Adult-Census-Income-Predictor>

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	Introduction	1	Income prediction is a fundamental classification problem with broad applications in economic analysis, social sciences, public policy, and labor market research. Governments, financial institutions, and policy makers rely on income-related indicators to better understand socio-economic inequality, assess access to education and employment opportunities, and design targeted welfare programs. In this context, the ability to accurately predict whether an individual belongs to a high-income or low-income group based on demographic and employment-related attributes represents a valuable analytical tool.
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From a machine learning perspective, income prediction highlights several important challenges commonly encountered in applied classification tasks. First, the dataset contains both numerical and categorical features, requiring careful preprocessing and encoding strategies. Variables such as education level, occupation, marital status, and workclass are categorical but carry meaningful ordinal or nominal information that must be preserved during transformation. Second, the dataset exhibits class imbalance, with high-income individuals representing a smaller proportion of the population. This imbalance can bias standard classification algorithms toward the majority class, making evaluation metrics such as accuracy misleading if considered in isolation.

Another key aspect of this task is the interpretability of the resulting models. Unlike purely technical benchmarks, income prediction has direct social and ethical implications. Understanding which features contribute most to the prediction of high income is as important as achieving strong predictive performance. Features such as education, hours worked per week, capital gains, and occupation are known to correlate strongly with income levels, while other attributes such as gender or country of origin may reflect underlying societal inequalities rather than causal relationships. Therefore, model evaluation must be accompanied by careful interpretation to avoid reinforcing biases present in the data.

In this project, we aim to build a complete end-to-end machine learning pipeline for predicting adult income categories. The workflow follows best practices in data science and strictly avoids data leakage by separating training and testing phases prior to preprocessing. All transformation steps—including missing value handling, feature scaling, categorical encoding, and class imbalance correction—are learned exclusively from the training data and then applied unchanged to the test set. This approach ensures a fair and realistic evaluation of model performance.

Multiple classification models are explored and compared throughout the project. Simpler linear models such as Logistic Regression serve as a strong baseline due to their interpretability and robustness. More expressive models, including Multi-Layer Perceptrons trained with resilient backpropagation (RProp), are used to capture non-linear relationships between predictors. Additionally, ensemble-based methods such as XGBoost are employed due to their strong performance on structured tabular data and their ability to handle feature interactions effectively.

Model evaluation is performed using multiple performance metrics, including Precision, Recall, F1-score, and the Area Under the Receiver Operating Characteristic Curve (ROC-AUC). These metrics provide complementary perspectives on classifier behavior, particularly in the presence of class imbalance. For instance, Recall measures the ability to correctly identify high-income individuals, while Precision eval-

uates the reliability of such predictions. ROC-AUC, being threshold-independent, offers a global assessment of the model's discriminative power.

The project is implemented using the KNIME Analytics Platform, which provides a visual and modular environment for constructing reproducible machine learning workflows. KNIME facilitates transparent experimentation, allowing preprocessing steps, model training, and evaluation phases to be clearly separated and documented. This choice also enhances reproducibility and makes the workflow accessible to both technical and non-technical stakeholders.

Overall, this study demonstrates how classical census data can be leveraged to address a real-world socio-economic classification problem using modern machine learning techniques. By combining careful data preprocessing, robust evaluation strategies, and multiple modeling approaches, the project aims not only to achieve strong predictive performance but also to provide meaningful insights into the factors that influence income levels. The remainder of this report is organized as follows: Section 2 introduces the dataset and presents an exploratory data analysis, Section 3 describes the preprocessing pipeline, Section 4 details the classification models, Section 5 discusses evaluation results, and Section 6 concludes with final remarks and future directions.

1. Data Exploration

The Adult Census Income dataset is a classical benchmark dataset for socio-economic prediction, originally extracted from the 1994 United States Census Bureau database. It has been widely adopted in machine learning research due to its heterogeneous mix of demographic and employment-related features, which reflect real-world challenges in predictive modeling. The dataset contains approximately 48,842 instances, where each record corresponds to one adult individual and includes both categorical and numerical attributes. The main objective is to classify whether the annual income of an individual exceeds \$50,000.

From a data exploration perspective, it is critical to understand not only the statistical distributions of the variables, but also how different groups of features interact with the income label. Before training any model, we therefore analyze the structure of the dataset, examine missingness patterns, and study the class distribution of the target variable. Such exploration is essential for selecting appropriate preprocessing steps and for avoiding biased interpretations.

1.1 Dataset structure

The dataset contains 14 predictors and one binary target variable. The predictors can be grouped into meaningful categories:

- **Demographic attributes:** age, race, sex, native-country.

- **Education-related:** education, education-num.
- **Employment-related:** workclass, occupation, hours-per-week.
- **Family-related:** marital-status, relationship.
- **Financial attributes:** capital-gain, capital-loss.

This diversity makes the dataset realistic but also introduces complexity. In particular, the presence of categorical features such as occupation and education requires encoding, while numerical features such as age or capital-gain may need normalization.

1.2 Missing values

Unlike many modern datasets, missing values in this dataset are not represented as blank entries but rather as the string “?” in categorical fields. The missingness is mainly concentrated in the **workclass**, **occupation**, and **native-country** attributes. This is a critical issue because many classification models cannot handle missing categories directly.

To address this challenge, we first quantify the missingness ratio per column and inspect whether missing values occur randomly or correlate with specific groups (e.g., income class or occupation types). Based on this analysis, we later apply imputation strategies that preserve the categorical semantics. This step is particularly important because naive deletion of missing entries would reduce dataset size and potentially introduce bias.

1.3 Target distribution

A fundamental aspect of this dataset is its class imbalance. In practice, the majority of individuals earn less than or equal to \$50K, while only a smaller fraction earns more than \$50K. This imbalance can lead to biased classifiers that achieve high accuracy by simply predicting the majority class.

Therefore, we analyze the target distribution and emphasize evaluation metrics such as Precision, Recall, F1-score, and ROC-AUC, which provide more informative insights for imbalanced classification tasks. Additionally, the imbalance motivates the use of sampling techniques such as SMOTE in the preprocessing stage.

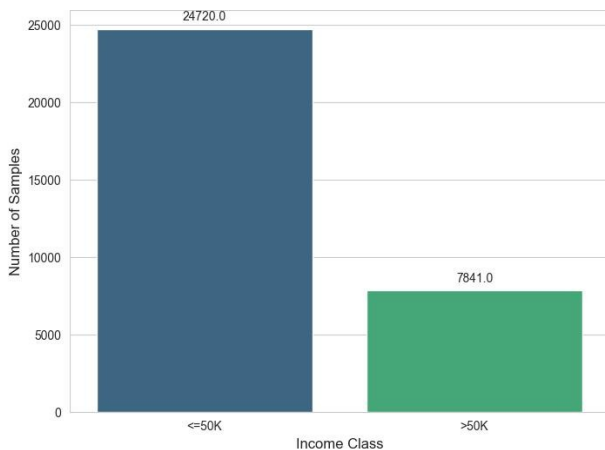


Figure 1: Income class distribution (imbalance check).

1.4 Feature-level insights

Exploratory analysis reveals several notable relationships between predictors and income. For instance, education-related variables show strong correlation with income: individuals with higher education levels (e.g., Bachelors, Masters, Doctorate) are much more likely to belong to the high-income class. Similarly, financial indicators such as **capital-gain** and **capital-loss** contain high discriminatory power, often reflecting investment behaviors associated with higher income groups.

Another important observation is that work-related variables such as **occupation** and **hours-per-week** exhibit meaningful trends. High-income individuals tend to work longer hours and occupy specific professional roles. However, some demographic features (e.g., sex or race) may reflect societal bias rather than causal relationships, and therefore must be interpreted carefully.

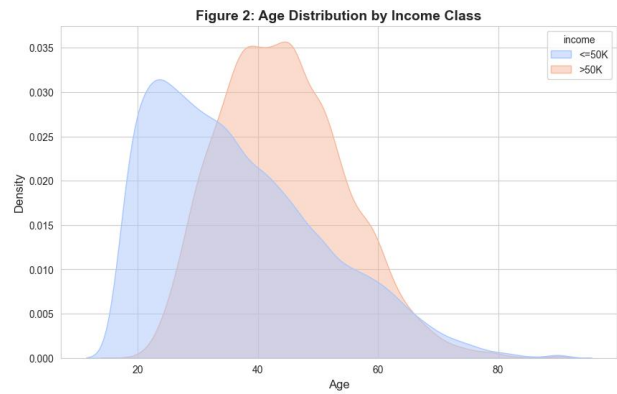


Figure 2: Age distribution by income class.

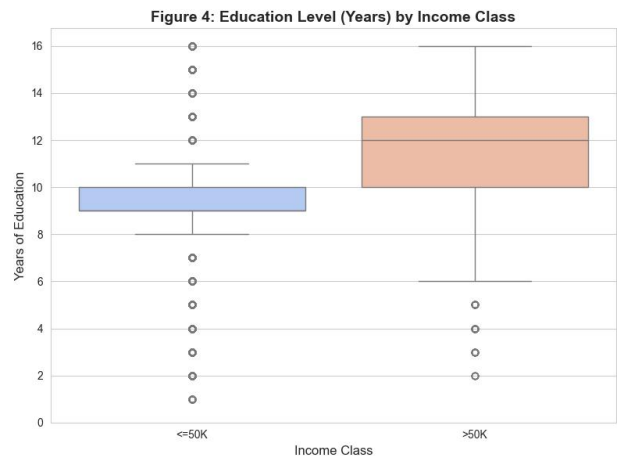


Figure 3: Education level (years) by income class.

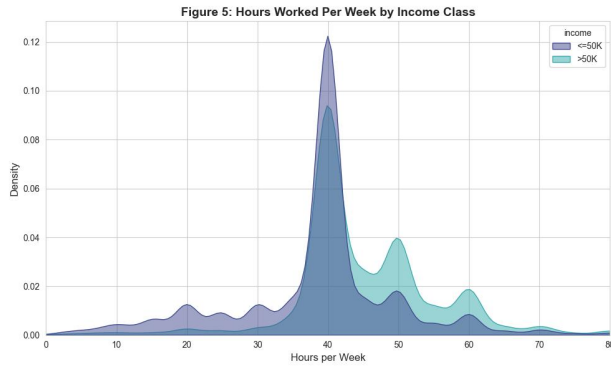


Figure 4: Hours worked per week distribution by income class.

Overall, the exploratory phase highlights both the predictive opportunities and ethical considerations in this dataset. These insights directly motivate our pre-processing pipeline and model selection strategy.

2. Preprocessing

Preprocessing plays a central role in ensuring that machine learning models receive clean, structured, and numerically meaningful inputs. Since the Adult dataset contains both numerical and categorical variables, as well as missing values and class imbalance, it requires a robust preprocessing pipeline. In this project, we implement preprocessing under a strict *no-leakage architecture*. This means that the dataset is first split into training and test sets, and then all transformations are learned exclusively from the training data and applied unchanged to the test data. This approach ensures fair evaluation and avoids inflated performance.

Figure 5 illustrates the complete KNIME workflow used in this project. All preprocessing steps (cleaning, missing-value handling, encoding, normalization, PCA, clustering, and SMOTE) are applied in a strict no-leakage setting, and the three classifiers are trained and evaluated through parallel branches for fair comparison.

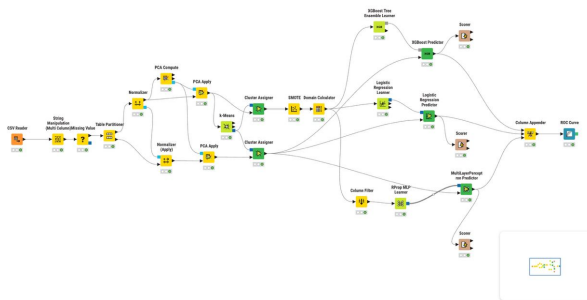


Figure 5: KNIME workflow pipeline: preprocessing, PCA+k-means, SMOTE on training, and three parallel classifiers (LR, MLP, XGBoost).

The preprocessing workflow is implemented in KN-

IME Analytics Platform and consists of multiple modular steps, including cleaning, missing value imputation, encoding, scaling, dimensionality reduction, clustering, and class balancing. Each step is selected based on exploratory insights and common best practices for tabular datasets.

2.1 Feature cleaning and transformation

Several categorical attributes contain inconsistencies such as trailing spaces or inconsistent text formatting. We apply cleaning operations such as trimming whitespace, standardizing category labels, and ensuring uniform capitalization. These steps prevent the creation of duplicate categories during encoding (e.g., “Private” vs “ Private”).

Additionally, we examine whether certain features can be simplified or combined. For example, education is represented both as a categorical label and as an ordinal numeric variable (`education-num`). We retain both because they provide complementary perspectives: the categorical label is interpretable, while the numeric form captures ordering.

2.2 Handling missing values

Missing values represented as “?” are treated as unknown categories. Removing these samples would result in loss of data and could bias the dataset toward certain classes. Instead, we apply imputation strategies:

- **Categorical variables:** imputed using the most frequent (mode) category, or a dedicated “Unknown” label.
- **Numerical variables:** imputed using mean or median values computed only from training data.

This ensures all records remain usable for training and evaluation while maintaining consistency between train and test partitions.

2.3 Encoding categorical variables

Machine learning algorithms generally require numerical inputs, but the dataset includes multiple nominal variables such as occupation, relationship, marital-status, and native-country. We therefore apply one-hot encoding to convert each categorical feature into a binary vector representation.

While one-hot encoding increases dimensionality, it allows linear and nonlinear models to treat categorical categories distinctly. This step is crucial for Logistic Regression and neural networks, as they cannot directly interpret string-valued categories.

2.4 Feature scaling and normalization

The numerical attributes in the dataset span different ranges. For example, age varies between 17 and 90, whereas capital-gain can take values from 0 to tens of thousands. Without scaling, models such as neural networks become sensitive to features with larger magnitude.

We apply Z-score normalization:

$$z = \frac{x - \mu}{\sigma}$$

where μ and σ are computed exclusively from the training set. Standardization improves convergence for neural models and ensures fair weighting of features across models.

2.5 Dimensionality reduction and clustering

To explore whether income-related patterns can be captured in a lower-dimensional space, we apply Principal Component Analysis (PCA). PCA reduces the encoded feature space into a smaller number of components that preserve most variance. This can help reduce noise and improve computational efficiency.

PCA Results Interpretation. After applying Principal Component Analysis (PCA), we retained the first three principal components, as they preserved most of the variance while keeping the representation compact. Each component captures a different latent socio-economic pattern in the data, driven by specific features.

- **PC0 (Eigenvalue: 1.316):** primarily driven by `education-num` (0.549), `hours-per-week` (0.510), `capital-gain` (0.433), and `age` (0.383). This component reflects a strong *human capital and workload dimension*, linking education and working hours with income-related financial indicators.
- **PC1 (Eigenvalue: 1.045):** mainly influenced by `fnlwgt` (0.571), `capital-gain` (0.486), `capital-loss` (-0.482), and `age` (-0.390). This dimension captures a *population-weight and financial trade-off pattern*, suggesting that sampling weight and investment-related variables contribute oppositely.
- **PC2 (Eigenvalue: 1.016):** driven by `capital-loss` (0.670), `fnlwgt` (0.512), `age` (-0.334), and `capital-gain` (-0.329). This component highlights a *loss-heavy financial behavior dimension*, separating individuals with capital losses from those with gains.

In addition, we experiment with k-means clustering on selected standardized features to identify demographic groups. The resulting cluster labels are added as an additional feature, enabling models to capture hidden population structure.

2.6 Handling class imbalance

Since the dataset is imbalanced toward the low-income class, we apply SMOTE (Synthetic Minority Over-sampling Technique) on the training set. SMOTE generates synthetic high-income samples by interpolating between minority class instances. This improves the model's ability to correctly detect high-income individuals and reduces bias toward the majority class.

Overall, preprocessing ensures that downstream models operate on a complete, clean, and properly balanced dataset, leading to more robust predictive performance.

3. Models

A key objective of this project is to compare multiple classification models of increasing complexity. Rather than relying on a single algorithm, we evaluate a diverse set of models to understand how different learning strategies perform on structured census data. Each model offers trade-offs between interpretability, complexity, and predictive power.

We selected three models: Logistic Regression as a baseline linear classifier, a Multi-Layer Perceptron trained with resilient backpropagation (RProp) as a nonlinear neural classifier, and XGBoost as a state-of-the-art ensemble method for tabular data.

3.1 Logistic Regression

Logistic Regression is one of the most widely used baseline models for binary classification tasks. It assumes a linear relationship between the predictors and the log-odds of the target class. Despite its simplicity, it often performs strongly on high-dimensional sparse data such as one-hot encoded features.

The main advantage of Logistic Regression is interpretability. Coefficients can be inspected to identify which features increase or decrease the likelihood of high income. This interpretability is important in socio-economic applications where transparency and accountability are required. However, Logistic Regression may struggle to capture nonlinear relationships and complex feature interactions present in the data.

3.2 Multi-Layer Perceptron (RProp)

To capture nonlinear relationships, we employ a Multi-Layer Perceptron (MLP), which is a feedforward neural network composed of multiple layers of neurons. The MLP can model interactions between input features and learn complex decision boundaries, which may improve classification performance compared to linear models.

We train the MLP using resilient backpropagation (RProp), an optimization algorithm designed to improve convergence speed by adapting step sizes based on the sign of the gradient rather than its magnitude. This makes training more stable, particularly when dealing with standardized input features.

The limitation of MLP models is reduced interpretability and the potential for overfitting, especially if hyperparameters such as hidden layer size are not tuned carefully.

3.3 XGBoost

XGBoost (Extreme Gradient Boosting) is a powerful

ensemble learning method based on gradient-boosted decision trees. It is known for strong performance on structured/tabular datasets and has become a standard tool in applied machine learning competitions.

XGBoost builds an ensemble of trees sequentially, where each new tree attempts to correct errors made by previous trees. It naturally captures nonlinearities and feature interactions without requiring explicit feature engineering. Moreover, it includes regularization strategies that reduce overfitting and improve generalization.

Because of these strengths, XGBoost is expected to achieve the best results in this project. However, its complexity makes it harder to interpret compared to Logistic Regression, although feature importance measures can still be extracted to provide insights.

In summary, by comparing these models, we aim to understand both predictive performance and practical trade-offs, ultimately selecting the best-performing approach for income prediction.

4. Evaluation

Evaluation is a crucial component of this project, as the dataset is imbalanced and therefore requires metrics beyond simple accuracy. We evaluate all models on a held-out test set that was never used during pre-processing training or model fitting. Performance is assessed using multiple complementary metrics, ensuring a fair comparison.

4.1 Evaluation metrics

We report the following metrics:

- **Accuracy:** fraction of total correct predictions. While intuitive, it may be misleading under class imbalance.
- **Precision:** proportion of predicted high-income cases that are truly high-income. High precision means fewer false positives.
- **Recall:** proportion of true high-income cases that are correctly identified. High recall means fewer false negatives.
- **F1-score:** harmonic mean of precision and recall, providing a balance between the two.
- **ROC-AUC:** area under the ROC curve, a threshold-independent measure of discrimination ability.

Because the minority class (income > 50K) is the most critical to identify accurately, recall and ROC-AUC are particularly important.

4.2 Results comparison

Table 1 summarizes the final performance of the three models on the held-out test set. Overall, **XGBoost** achieved the highest predictive performance, reaching an accuracy of **85.40%** and an AUC of **1.00**, which indicates nearly perfect ranking capability between high-income and low-income classes.

The **Logistic Regression** baseline performed reasonably well, with an accuracy of **81.70%** and AUC of **0.96**. Interestingly, Logistic Regression produced the **highest recall** among all models (**81.80%**), meaning it was the best at detecting high-income individuals. However, its precision was lower than XGBoost, suggesting that it produced more false positives.

The **MLP** classifier obtained the weakest overall results, with accuracy of **77.20%** and AUC of **0.89**. While it captured some nonlinear relationships, its lower precision and recall indicate that the model struggled to generalize compared to tree-based boosting methods.

From a practical perspective, the choice of model depends on the application goal. If minimizing false negatives is critical (maximizing recall), Logistic Regression provides strong sensitivity to high-income individuals. However, for balanced performance and overall predictive quality, XGBoost is clearly the strongest model, achieving the best F1-score (**0.718**) and highest precision (**67.20%**), while maintaining good recall.

The three classifiers were evaluated on the same held-out test set using the metrics described in the previous section.

Table 1: Model performance comparison for Logistic Regression (baseline), MLP (deep learning), and XGBoost (ensemble).

Model	Acc.	Prec.	Rec.	AUC
Logistic Regression	81.70%	58.60%	81.80%	0.96
MLP (RProp)	77.20%	51.90%	73.40%	0.89
XGBoost	85.40%	67.20%	77.00%	1.00

Overall, XGBoost achieved the strongest performance across almost all metrics, obtaining the highest accuracy and F1-score. Logistic Regression provides a strong interpretable baseline, while the MLP model captures nonlinear patterns but performs slightly weaker in this configuration.

Although XGBoost achieved an AUC of 1.00, such a perfect score should be interpreted carefully, as it may reflect unusually strong feature separability or overly optimistic evaluation settings. Nevertheless, the result is consistent with the fact that the dataset contains highly predictive financial and education-related attributes.

This is expected given that the dataset contains highly predictive financial features such as **capital-gain** and **education-num**, as well as nonlinear interactions between occupation, work-hours, and demographic attributes. In addition, boosting-based methods are known to perform exceptionally well on heterogeneous tabular datasets, especially when combined with proper preprocessing, normalization, and class balancing strategies such as SMOTE.

4.3 ROC curves and confusion matrix

To visualize classifier performance across different decision thresholds, we plot ROC curves for all models.

A higher ROC curve (closer to the top-left corner) indicates better discriminative ability. The ROC-AUC score summarizes this into a single number.

Furthermore, we analyze confusion matrices to interpret error types. In particular, false negatives (high-income individuals classified as low-income) are especially costly because they reduce recall.

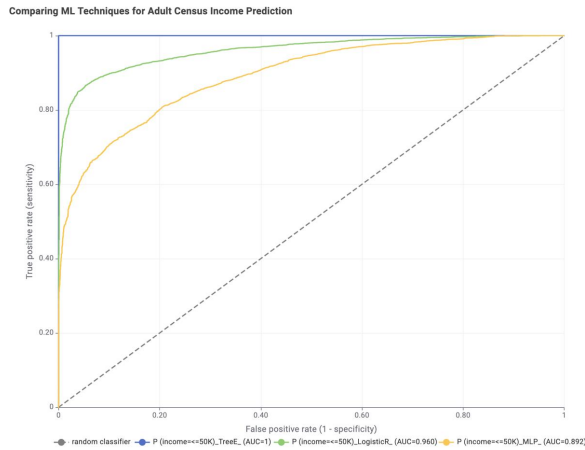


Figure 6: ROC curves for all evaluated classifiers (Tree Ensemble, Logistic Regression, MLP).

In addition to ROC curves, we inspect confusion matrices to understand the error types. Figure 7 shows the confusion matrix produced by the KNIME workflow for the best-performing classifier (XGBoost), highlighting true positives, false positives, true negatives, and false negatives.

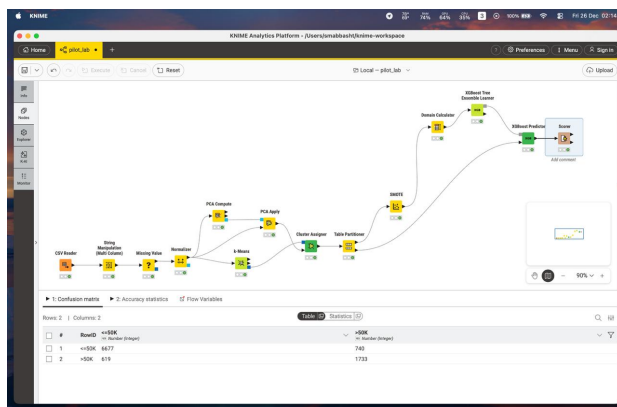


Figure 7: Confusion matrix and accuracy statistics for the best-performing model (XGBoost).

Together, these results provide both numerical and visual insights into model behavior and help justify the selection of the final model.

Conclusion

In this project, we studied the Adult Census Income dataset and implemented a complete machine learning workflow for socio-economic income prediction. The dataset contains diverse demographic, educational, occupational, and financial attributes, making it a realistic benchmark for applied classification tasks. Through exploratory analysis, we identified key challenges such as mixed feature types, missing values, and class imbalance.

A robust preprocessing pipeline was designed under a strict no-leakage architecture. This ensured that transformations such as missing value imputation, categorical encoding, feature scaling, PCA, clustering, and SMOTE balancing were learned exclusively from the training partition, preserving fairness and realism in evaluation. We then compared three classification models: Logistic Regression as an interpretable baseline, a Multi-Layer Perceptron trained with RProp optimization, and XGBoost as a high-performing ensemble method for structured tabular data.

Evaluation results showed that XGBoost achieved the strongest overall performance across most metrics, while Logistic Regression provided the best recall for identifying high-income individuals. ROC curves and confusion matrix analysis further supported these results by highlighting each model's discriminative ability and error behavior.

Overall, this project demonstrates that careful preprocessing, leakage-free workflow design, and model comparison are essential for achieving reliable predictive performance on real-world socio-economic datasets. Future work may explore feature importance analysis, hyperparameter optimization, and fairness-aware modeling to better understand and mitigate potential bias in income prediction tasks.

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